

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

Slot A

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks 50
Annexure A MCQ

Duration :60 Minutes

Code of Conduct:

I P. Akhila / 192472320 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Akhila

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Preprocessing – Normalization & Scaling

A 12-bit image has pixel value $r = 3000$.

The image is normalized to $[0,1]$ and then scaled to 8-bit $[0,255]$.

What is the final pixel value?

- A) 150
- B) 187
- C) 200
- D) 225

Ans: B) 187

2. Image Storage Calculation

An RGB image of size 2048×1024 uses 8 bits per channel. What is the storage in MB?

- A) 6 MB
- B) 8 MB
- C) 12 MB
- D) 24 MB

Ans: A) 6 MB

3. Sobel Edge Magnitude

At a pixel: ($G_x = 10$, $G_y = 24$)

Compute the gradient magnitude (approx).

- A) 20
- B) 25
- C) 26
- D) 30

Ans: C) 26

4. Sobel Edge Direction

For the same ($G_x = 10$, $G_y = 24$), compute edge direction in degrees.

- A) 22°
- B) 67°
- C) 78°
- D) 90°

Ans: B) 67°

5. Harris Corner Response $\det(M) =$

500, $\text{trace}(M) = 40$, $k = 0.04$

Compute Harris response R .

- A) 440
- B) 460
- C) 500
- D) 520

Ans: A) 440

6. Harris Corner vs Edge Classification Harris

matrix eigenvalues: $\lambda_1 = 100$, $\lambda_2 = 5$

The pixel corresponds to:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans: B) Edge

7. Hough Transform – Line Parameter A

line in Hough space: $\rho = 50$, $\theta = 90^\circ$

Which line is represented?

- A) Horizontal line $y = 50$
- B) Vertical line $x = 50$
- C) Diagonal
- D) Random line

Ans: B) Vertical line $x = 50$

8. SIFT Scale Adjustment

Original keypoint $\sigma = 1.2$, image scaled by factor 2.

New keypoint scale = ?

- A) 1.2
- B) 2.4
- C) 0.6
- D) 3.2

Ans: B) 2.4

9. PCA Variance Calculation

Eigenvalues of a dataset: 50, 25, 15, 10

How many principal components retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4
- D) 5

Ans: B) 3

10. PCA Dimensionality Reduction Impact

Original feature vector: 200 dimensions
Reduced via PCA: 40 dimensions

Percentage reduction = ?

- A) 70%
- B) 75%
- C) 80%

Ans: C) 80%

11. Preprocessing – Normalization Scenario

A drone captures an 12-bit image with pixel value $r = 2500$. Before feature extraction, it is normalized to $[0,1]$.

What is the normalized value?

- A) 0.55
- B) 0.61
- C) 0.65
- D) 0.72

Ans: B) 0.61

12. Image Storage Scenario

A grayscale image: 2048×1024 , 16-bit per pixel. You convert it to 8-bit for faster processing.

What is the percentage reduction in memory usage?

- A) 25%
- B) 50%

C) 60%

D) 75%

Ans: B) 50%

13. Edge Detection Scenario

A Sobel operator detects gradients at a pixel: ($G_x = 12$, $G_y = 16$).

Compute the gradient magnitude.

A) 18

B) 20

C) 24

D) 28

Ans: B) 20

14. Edge Direction Scenario

For the same pixel above, compute gradient direction θ .

A) 36.9°

B) 45°

C) 53.1°

D) 60°

Ans: A) 36.9°

15. Harris Corner Detection Scenario

Region M parameters: $\det(M) = 600$, $\text{trace}(M) = 30$, $k = 0.05$

Compute Harris response R.

A) 540

B) 555

C) 570

D) 600

Ans: B) 555

16. Harris Corner Classification Scenario

Eigenvalues at a pixel: $\lambda_1 = 100$, $\lambda_2 = 90$

The pixel is most likely:

A) Flat region

B) Edge

- C) Corner
- D) Noise

Ans: C) Corner

17. Hough Transform Scenario

Two points in an image produce intersecting curves in Hough space at ($\rho = 80$, $\theta = 60^\circ$).

What does this indicate?

- A) Detected corner
- B) Line in image space
- C) Circular feature
- D) Noise

Ans: B) Line in image space

18. SIFT Scale Scenario

A SIFT keypoint appears at scale $\sigma = 2.0$.

The image is downsampled by a factor of 0.5.

What is the new keypoint scale?

- A) 0.5
- B) 1.0
- C) 2.0
- D) 4.0

Ans: B) 1.0

19. PCA Variance Scenario

A dataset has eigenvalues: 70, 15, 10, 5

How many principal components retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4

Ans: B) 3

20. PCA Dimensionality Reduction Scenario

Original feature vector: 150 dimensions
Reduced via PCA: 25 dimensions

What is the percentage reduction?

- A) 70%
- B) 75%
- C) 83%
- D) 85%

Ans: C) 83%

21. Preprocessing – Normalization & Scaling

A 12-bit satellite image has a pixel intensity $r = 3500$.
It is normalized to $[0,1]$ and then scaled to 8-bit $[0,255]$ for processing.

Question: What is the final 8-bit pixel value?

- A) 190
- B) 210
- C) 218
- D) 225

Ans: C) 218

22. Image Storage Analysis

A 1024×1024 RGB image with 8-bit per channel is converted to grayscale 8-bit.

Question: What is the percentage reduction in storage?

- A) 33%
- B) 50%
- C) 66%
- D) 75%

Ans: C) 66%

23. Edge Detection – Gradient Magnitude A pixel

has Sobel gradients: ($G_x = 14$, $G_y = 48$)

Question: Compute the gradient magnitude.

- A) 46
- B) 50
- C) 50.5
- D) 50.99

Ans: B) 50

24. Edge Direction

For the same pixel: ($G_x = 14$, $G_y = 48$)

Question: Compute edge orientation in degrees.

- A) 70°
- B) 73°
- C) 75°
- D) 77°

Ans: C) 75°

25. Harris Corner Response

Harris parameters: $\det(M) = 400$, $\text{trace}(M) = 35$, $k = 0.05$

Question: Compute Harris response R.

- A) 320
- B) 350
- C) 378
- D) 400

Ans: No exact option

26. Harris Corner Classification Eigenvalues

of M: $\lambda_1 = 130$, $\lambda_2 = 120$

Question: Pixel type?

- A) Flat
- B) Edge
- C) Corner
- D) Noise

Ans: C) Corner

27. Hough Transform – Line Detection

Two points in image space generate intersecting curves in Hough space at ($\rho = 100$, $\theta = 45^\circ$).

Question: What does this indicate in the image?

- A) A line passes through both points
- B) A corner formed by lines
- C) Circular feature
- D) Random noise

Ans: A) A line passes through both points

28. SIFT Scale Scenario

A SIFT keypoint at scale $\sigma = 1.5$ is observed. The image is scaled by $0.25\times$.

Question: New keypoint scale?

- A) 0.375
- B) 0.75
- C) 1.5
- D) 6

Ans: A) 0.375

29. PCA Variance Retention

Eigenvalues: 60, 20, 10, 5, 5

Question: Minimum principal components to retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4
- D) 5

Ans: B) 3

30. PCA Dimensionality Reduction

Original vector dimension = 180 After
PCA = 30 components

Question: Percentage reduction?

- A) 70%
- B) 75%
- C) 83%
- D) 85%

Ans: C) 83%

31. Preprocessing Method Selection

A remote sensing image contains high sensor noise and uneven illumination. You need to enhance edges for feature extraction.

Which preprocessing combination is most appropriate?

- A) Histogram equalization + median filtering
- B) Gaussian smoothing + normalization
- C) Median filtering + adaptive histogram equalization

D) Low-pass filtering + quantization

Ans: C) Median filtering + adaptive histogram equalization

32. Image Representation Trade-Off

A 3D medical scan (1024×1024) can be stored as 16-bit grayscale or 8-bit with histogram equalization.

Which representation maximizes detail and reduces storage without losing features?

- A) 16-bit grayscale
- B) 8-bit grayscale with HE
- C) 8-bit grayscale without HE
- D) RGB 24-bit

Ans: B) 8-bit grayscale with HE

33. Edge Detection Algorithm Choice

You are analyzing a satellite image with low contrast edges and some noise.

Which edge detector is most suitable?

- A) Sobel
- B) Prewitt
- C) Canny
- D) Roberts

Ans: C) Canny

34. Harris vs FAST Corner Selection

You need to detect corners for real-time object tracking.

Which method is preferred for speed without losing accuracy?

- A) Harris
- B) FAST
- C) SIFT
- D) Hough Transform

Ans: B) FAST

35. Hough Transform Line Detection Evaluation

Two line detection methods are applied on a road image:

- Method 1: ρ -resolution = 1 pixel, θ -resolution = 1° •
- Method 2: ρ -resolution = 5 pixels, θ -resolution = 5°

Which method balances accuracy and computational cost best?

- A) Method 1
- B) Method 2
- C) Both equal
- D) Use Canny edges only

Ans: B) Method 2

36.. SIFT vs SURF Evaluation

You are matching keypoints between two images with scale and rotation differences.

Which method ensures robust matching with lower computation?

- A) SIFT
- B) SURF
- C) Harris + SSD
- D) Hough Transform

Ans: B) SURF

37. PCA Component Selection

A dataset has eigenvalues: 70, 20, 10, 5, 5

To retain $\geq 95\%$ variance for classification:

- A) 2 components
- B) 3 components
- C) 4 components
- D) 5 components

Ans: C) 4 components

38. Feature Extraction Evaluation

You have a noisy industrial inspection image. You want to detect straight edges of objects.

Which combination is optimal?

- A) Sobel + Hough Transform
- B) Canny + Harris Corner
- C) Median Filter + SIFT
- D) Gaussian Blur + PCA

Ans: A) Sobel + Hough Transform

39. Preprocessing for SIFT Matching

Two aerial images have different brightness and contrast.

What preprocessing ensures scale-invariant feature matching works best?

- A) Histogram equalization + normalization
- B) Median filtering + low-pass filter
- C) Gaussian smoothing only
- D) No preprocessing

Ans: A) Histogram equalization + normalization

40. PCA Dimensionality Evaluation

A feature vector of 200 dimensions is reduced using PCA to 30 components, retaining $\approx 90\%$ variance.

You need to classify images quickly.

Which evaluation is correct?

- A) Reduction sufficient for speed and accuracy
- B) Reduce to 10 components for maximum speed
- C) Keep 200 components for highest accuracy
- D) Use SIFT only

Ans: A) Reduction sufficient for speed and accuracy

41. Preprocessing – Normalization & Decision

A 12-bit satellite image has a pixel $r = 3200$.

You normalize it to $[0,1]$ and scale to 8-bit $[0,255]$.

Question: What is the final pixel value?

- A) 198
- B) 199
- C) 200

Ans: B) 199

42. Image Storage Trade-Off

A 1024×1024 RGB image (8-bit per channel) is converted to grayscale 8-bit to save storage.

Question: What is the percentage reduction?

- A) 33%
- B) 50%
- C) 66.7%
- D) 75%

Ans: C) 66.7%

43. Sobel Edge Magnitude Evaluation

A pixel has gradients ($G_x = 12$, $G_y = 16$).

Question: Compute gradient magnitude and evaluate which pixel is stronger than another pixel with ($G_x = 10$, $G_y = 18$).

- A) 20; weaker
- B) 20; stronger
- C) 21; weaker
- D) 21; stronger

Ans: A) 20; weaker

44. Harris Corner Response Analysis $\det(M)=400$, $\text{trace}(M)=30$, $k=0.04$

Question: Compute R and evaluate whether it's more likely a corner than a pixel with $\det(M)=350$, $\text{trace}(M)=25$, $k=0.04$.

- A) $R=364$; less likely
- B) $R=364$; more likely
- C) $R=362$; less likely
- D) $R=362$; more likely

Ans: B) $R=364$; more likely

45. Harris Eigenvalue Evaluation

Eigenvalues of M at a pixel: $\lambda_1=120$, $\lambda_2=80$

Question: Classify pixel as flat, edge, or corner numerically.

- A) Flat
- B) Edge
- C) Corner
- D) Noise

Ans: C) Corner

46. Hough Transform Voting Numerical

Two points in image space: (100,50) and (200,100)

Each generates sinusoidal curves in Hough space. They intersect at ($\rho=70$, $\theta=45^\circ$).

Question: What does this indicate?

- A) Line passes through both points
- B) Corner
- C) Noise

D) Circle

Ans: A) Line passes through both points

47. SIFT Keypoint Scale Numerical Evaluation

Original keypoint $\sigma=1.6$; image scaled $0.5\times$

Question: New σ value?

A) 0.5

B) 0.8

C) 1.6

D) 3.2

Ans: B) 0.8

48. PCA Variance Contribution Evaluation

Eigenvalues: 60, 20, 10, 5, 5

Question: Minimum principal components to retain $\geq 90\%$ variance?

A) 2

B) 3

C) 4

D) 5

Ans: B) 3

49. PCA Dimensionality Reduction Numerical Evaluation

Original feature dimension = 200 After

PCA = 30 components

Question: Evaluate percentage reduction.

A) 70%

B) 75%

C) 80%

D) 85%

Ans: D) 85%

50. Preprocessing Impact Evaluation

Two 8-bit images:

- Image A has uniform illumination

- Image B has low contrast and brightness differences

You apply histogram equalization to both.

Question: Evaluate which image benefits more numerically in contrast enhancement.

- A) Image A
- B) Image B
- C) Both equal
- D) Neither

Ans: B) Image B